

Meghana Narayana

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Summary

Full-stack engineer with 3+ years building enterprise **ASP.NET Core** APIs and **Angular** dashboards in regulated healthcare at IQVIA and shipping **generative AI** features powered by GPT-4o and prompt engineering with **React** and **Node.js** at the United Nations. Bridges microservices architecture, RESTful API design, and **SQL Server** optimization across large-scale pharmaceutical datasets with modern AI product development, with production deployments across **Azure** and **AWS**.

Education

Master of Science, Computer Science | **Northeastern University**

Boston, MA Sep 2024 - Apr 2026

Bachelor of Engineering | **RV College of Engineering**

Bengaluru, India Aug 2017 - May 2021

Experience

United Nations

Software Engineer Co-op

Boston, MA · May 2025 - Aug 2025

- Built an interactive governance analytics dashboard in **React** and **TypeScript** with D3.js visualizations, enabling policy analysts to explore and compare multi-government datasets without manual data exports
- Engineered a **RAG** pipeline using **pgvector**, OpenAI Embeddings, and **GPT-4o**, enabling UN policymakers to dynamically synthesize multiple governance case studies into consolidated reports on demand
- Designed and optimized **Node.js Express** APIs with indexed multi-filter **PostgreSQL** queries on **AWS RDS**, achieving **sub-300ms** lookups across global governance datasets

IQVIA Healthcare

Software Development Engineer

India · Aug 2022 - Jul 2024

- **Architected** a decoupled **ASP.NET Core API** and **Angular** dashboard infrastructure on **Azure App Services**, scaling system capacity to support concurrent high-throughput data analysis for **10K+** enterprise users
- Engineered an **event-driven** pipeline with **Azure Functions** and **Service Bus** to ingest 10K+ daily clinical records, using dead-letter queues and retry handling to guarantee no record loss in a regulated data workflow
- **Migrated** a legacy MVC monolith into scalable ASP.NET Core Web APIs and modular, lazy-loaded Angular featuring secure **OAuth/JWT** role-based authorization to improve security and performance.
- **Resolved race condition** in concurrent drug data updates by implementing pessimistic locking, preventing data corruption across **5K+** records
- Implemented **Redis** distributed caching for frequently accessed drug reference data, decreasing database load by **65%** and reducing dashboard load times from 8s to under 3s
- **Refactored** 5K+ line legacy C# codebase with **Copilot**, cutting issue resolution time through bug detection and eliminating redundant logic

Associate Software Developer

India · Aug 2021 - Jul 2022

- Developed an **end-to-end clinical trial tracking application** using Angular and ASP.NET Core(C#), implementing an **EF Core**-backed state model to programmatically control workflow transition and eliminate manual review bottlenecks
- **Optimized API payload** via server-side pagination and tuned **SQL Server** stored procedures on 100K+ row datasets, improving dashboard render time by 70%
- **Cut production defects by 35%** by developing 50+ rigorous **xUnit/Moq tests** validating high-volume market intelligence data pipeline logic

Projects

Multi-Tenant Operations Dashboard | [Github](#) | **ASP.NET Core 9, Angular 19, SignalR, Redis, SQL Server, Azure, Docker**

- Architected a multi-tenant SaaS platform with JWT + RBAC and tenant-scoped Redis cache invalidation, cutting API latency by **85%**.
- Engineered predictive anomaly alerts in Angular using linear regression forecasts, pushing threshold warnings to operators via SignalR WebSockets before breaches occur.
- Designed SQL Server schema with stored procedures and indexed views; containerized with Docker and deployed via Azure DevOps pipelines.

RAG Document Assistant | [Github](#) | **Python, FastAPI, PydanticAI, pgvector, PostgreSQL, OpenAI, React, Docker**

- Built a full-stack AI assistant using **pgvector** and **PostgreSQL** that lets users query document collections and receive answers grounded in retrieved source text.
- Orchestrated a **PydanticAI** agent loop (gpt-4.1, 20 tool calls/turn) with a citation validator that retried upto **2x** on hallucinated citations.
- Optimized retrieval cost by skipping LLM calls on short queries and streaming answers to React over SSE with real-time citation rendering, deployed via Docker with JWT.

Snip - URL Shortener | [Github](#) | **Next.js, TypeScript, Node.js, Redis, MySQL, Prisma, Vercel, Queue**

- Engineered a middleware redirect engine with Redis cache-aside, eliminating DB queries on cache hits and keeping p95 latency below **5ms**.
- Decoupled click analytics via async QStash event queue, writing MySQL rows post-response without adding latency to the redirect path.
- Secured a public API with SHA-256 key auth and Redis sliding window rate limiting at 100 req/min, blocking abuse without adding overhead to cache-hit paths.

Skills

- **Languages & Frameworks:** TypeScript, JavaScript, Python, C#, ASP .NET Core, Node.js, React, Next.js, Angular, FastAPI, Express, Microservices
- **Databases & Caching:** SQL Server, PostgreSQL, Vector Databases, MySQL, Redis, Entity Framework, LINQ
- **AI & APIs:** OpenAI API, PydanticAI, RAG, Restful API, SSE, JWT, OAuth, MCP
- **Cloud & DevOps:** Azure (Certified, App Services, Functions, Service Bus), AWS (EC2, RDS, S3, SNS/SQS), Docker, Terraform, Git, CI/CD
- **Testing & Tools:** xUnit, Moq, TDD, Jest, Selenium, SonarQube, Visual Studio, Jira, Agile, Scrum, Confluence, Claude, Cursor